

DATA 505 (Statistics Using R): Exam 2

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze the `MA_schools` dataset from the `moderndive` package, which contains data on Massachusetts public high schools in 2017.

```
# Install moderndive package if not already installed
if (!require(moderndive)) {
  install.packages("moderndive")
}

# Load required packages
library(moderndive)
library(ggplot2)

# Load the MA_schools dataset
data(MA_schools)
```

Data Exploration (3 points)

1 (3 points)

Create a scatterplot showing the relationship between `perc_disadvan` (percent economically disadvantaged students) and `average_sat_math` (average SAT math score). Based on the plot, do you expect a positive or negative relationship?

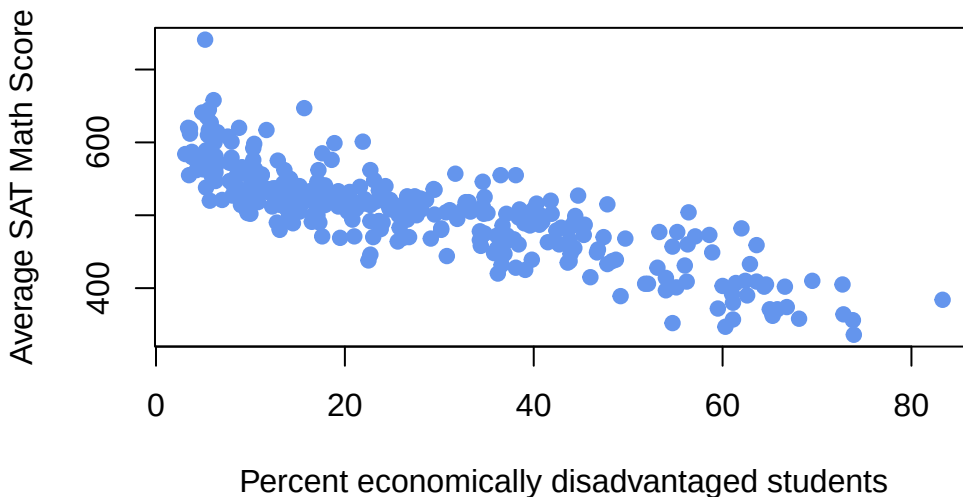
```
# YOUR CODE HERE

#View(MA_schools) #4 columns, 332 rows

#colnames(MA_schools)

plot(MA_schools$perc_disadvan, MA_schools$average_sat_math,
     main = "Relationship between economically disadvantaged Students and SAT Math scores",
     xlab = "Percent economically disadvantaged students",
     ylab = "Average SAT Math Score",
     pch = 19, col = "cornflowerblue")
```

Relationship between economically disadvantaged Students and SAT



Answer: Based on the plot, I would expect a negative relationship between the percentage of economically disadvantaged students and average SAT Math scores. The data points

form a downward trend from left to right, suggesting that as the percentage of economically disadvantaged students increases, the average SAT Math score tends to decrease.

Simple Linear Regression (12 points)

1 (4 points)

Fit a simple linear regression model predicting `average_sat_math` from `perc_disadvan`. Store the model as `model1` and display the summary.

```
# YOUR CODE HERE

model1 = lm(average_sat_math ~ perc_disadvan, data = MA_schools)

summary(model1)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-80.74	-21.26	-4.12	18.54	174.17

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	581.2811	3.2668	177.9	<2e-16 ***
perc_disadvan	-2.7798	0.1011	-27.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.54 on 330 degrees of freedom

Multiple R-squared: 0.6962, Adjusted R-squared: 0.6953

F-statistic: 756.2 on 1 and 330 DF, p-value: < 2.2e-16

2 (4 points)

Extract the estimated model coefficients (intercept and slope) from this fitted model object, and store them in a vector called `sat_effects`. Then report the intercept and slope below.

```
# YOUR CODE HERE
```

```
sat_effects = coef(model1) #or coefficients(model1) or model1$coefficients
sat_effects
```

```
(Intercept) perc_disadvan
581.281054      -2.779787
```

```
#Intercept: 581.281054, the predicted SAT Math score when perc_disadvan = 0
```

```
#Slope: -2.779787, the estimated change in SAT Math score for a one-unit increase in perc_d
```

Intercept: 581.281054, the predicted SAT Math score when perc_disadvan = 0

Slope: -2.779787, the estimated change in SAT Math score for a one-unit increase(1%) in perc_disadvan

3 (4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between the percentage of economically disadvantaged students and average SAT math scores?

Interpretation: For each 1% increase in economically disadvantaged students (perc_disadvan), the average SAT Math score is expected to decrease by about 2.78 points on average.

Making Predictions (8 points)

1 (4 points)

Use model1 to predict the average SAT math score for a school where 50% of students are economically disadvantaged. Create a data frame with this value and use the `predict()` function.

```
# YOUR CODE HERE
```

```
new_school = data.frame(perc_disadvan = 50)
```

```
predicted_sat_score = predict(model1, newdata = new_school)
```

```
predicted_sat_score
```

```
1
442.2917
```

Predicted average SAT math score: 442.2917

#The predicted average SAT math score is 442.29

2 (4 points)

Calculate and print out a 95% prediction interval for a single new school where 50% of students are economically disadvantaged.

```
# YOUR CODE HERE
```

```
predict(model1, newdata=new_school, interval = "prediction", level = 0.95)
```

```
      fit      lwr      upr
1 442.2917 376.0542 508.5292
```

We are 95% confident that the average SAT Math score for a new school with 50% of economically disadvantaged students will fall between 376 and 509 points.

Multiple Linear Regression (12 points)

1 (4 points)

Fit a multiple regression model predicting `average_sat_math` from both `perc_disadvan` and `size` (school size category). Store this as `model2` and display the summary.

```
# YOUR CODE HERE
```

```
model2 = lm(average_sat_math ~ perc_disadvan + size, data = MA_schools)
summary(model2)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan + size, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-81.342	-20.802	-3.872	18.211	167.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	588.1898	7.6067	77.325	<2e-16 ***
perc_disadvan	-2.7772	0.1063	-26.120	<2e-16 ***
sizemedium	-11.9134	7.5346	-1.581	0.115
sizelarge	-6.3615	6.9231	-0.919	0.359

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.5 on 328 degrees of freedom

Multiple R-squared: 0.6988, Adjusted R-squared: 0.696

F-statistic: 253.6 on 3 and 328 DF, p-value: < 2.2e-16

```
#this model accounts for school size to predict the average sat math score.
```

2 (4 points)

Compare the R-squared values of `model1` and `model2`. Which model explains more variance in average SAT math scores?

```
# YOUR CODE HERE
r2_model1 <- summary(model1)$r.squared
r2_model2 <- summary(model2)$r.squared

r2_model1
```

```
[1] 0.6961948
```

```
r2_model2
```

```
[1] 0.6987587
```

Answer: Model2 slightly explains more variance in average SAT math score. Model1 explains about 69.62% of the variance in the average SAT math scores while model2 explains about 69.88%. The difference is very small which shows that even after accounting for school size the R_squared did not increase much.

3 (4 points)

Use the `anova()` function to formally compare `model1` and `model2`. Does adding `size` significantly improve the model at the $\alpha = 0.05$ level?

```
# YOUR CODE HERE
```

```
anova(model1, model2)
```

Analysis of Variance Table

Model 1: `average_sat_math ~ perc_disadvan`

Model 2: `average_sat_math ~ perc_disadvan + size`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	330	371191				
2	328	368058	2	3132.6	1.3958	0.2491

Conclusion: Based on the ANOVA test ($p = 0.2491 > 0.05$) illustrating that adding `size` did not significantly improve the model at the $\alpha = 0.05$ level, indicating that ‘`perc_disadvan`’ alone explains most of the variance in average SAT math scores.